

CALCULUS 101 · ANSWER KEY - PROFESSOR COPY

Homework 2: How Much Piles Up?

Grading guide. Total 21 points. Answers in different words that mean the same thing get full credit; partial credit is suggested where it applies. Enter the final score in Professor mode on the recitation page.

Part A — Rate times time

1. Three pumps, three babies. Fill in the total each baby received. (Rate $\times$ hours. 2 points each: 1 for setting up the multiplication, 1 for the answer.) (6 pts)

Answer: A: $3 \times 24 = 72$ mL. B: $2.5 \times 8 = 20$ mL. C: $5 \times 12 = 60$ mL. (For B, accept $2 \times 8 = 16$ plus half of $8 = 4$, making 20.)

Part B — Draw the staircase

2. On the grid above, draw the pump's day: 2 mL/hr from hour 0 to 8, then 3 mL/hr from hour 8 to 16, then 4 mL/hr from hour 16 to 24. Shade under the steps. Then write the area of each rectangle and add them up for the day's total. (4 pts)

Answer: Three flat steps at heights 2, 3, 4 over 0-8, 8-16, 16-24 (2 pts; 1 pt off per wrong step). Areas: $2 \times 8 = 16$, $3 \times 8 = 24$, $4 \times 8 = 32$ (1 pt). Total 72 mL (1 pt).

Part C — The In-and-Out sheet

3. A real-style I&O sheet. IN: the baby had 8 tube feeds of 18 mL each. OUT: four diapers weighed 32 g, 28 g, 35 g, and 30 g; a dry diaper weighs 20 g. Fill in the totals. (1 gram of wet weight = 1 mL of urine.) (4 pts)

Answer: IN: $8 \times 18 = 144$ mL (1 pt). Diaper 1: $32 - 20 = 12$ mL (1 pt). Diapers 2-4: 8, 15, 10 mL (1 pt). All four: $12 + 8 + 15 + 10 = 45$ mL (1 pt).

Part D — Per kilogram

4. A baby weighs 1.2 kg and got 180 mL of milk today. What is that in mL/kg/day? Is she at 'full feeds' (about 150-160)? (2 pts)

Answer: $180 \div 1.2 = 150$ mL/kg/day (1 pt). Yes — right at full feeds (1 pt).

Part E — Flashbacks

5. From NICU Tutor: what are trophic feeds for? (1 pt)

Answer: Waking up an immature gut gently and letting it practice — not calories. Tiny amounts, a few mL a day.

6. From last week's Calculus: a baby went from 1,000 g to 1,140 g in 7 days. Rate of change? (1 pt)

Answer: $140 \div 7 = 20$ g/day.

Part F — Think like a doctor

7. The pump was set to 3 mL/hr all day, and the I&O sheet says 72 mL. But you find out the pump was paused for 4 hours during a procedure and nobody wrote it down. What did the baby really get? Show it on a rate graph in words (what happened to the rectangle?), and explain why the difference matters to the team. (3 pts)

Answer: Real total: $3 \times 20 = 60$ mL, not 72 (1 pt). On the graph, the rate line dropped to zero for 4 hours — a $3 \times 4 = 12$ mL notch cut out of the rectangle (1 pt). It matters because the baby got 12 mL fewer calories and fluid than the team thinks; if they plan tomorrow's feeds, fluids, or growth expectations from the wrong total, the baby can fall behind or the fluid balance can be misjudged — the sheet must match what actually happened (1 pt).

HOME LAB (optional, not graded): The dripping cup

You need: a paper or plastic cup, a pin or sharp pencil, a measuring cup with mL marks, a timer, a sink or tray

1. Poke a small hole in the bottom of the cup. Cover it with a finger, fill the cup with water, and hold it over the measuring cup.
2. Let go and time exactly 1 minute. Read the mL. That's your rate in mL per minute. Write it down.
3. PREDICT: at that rate, how much water will be in the measuring cup after 5 minutes? ($\text{Rate} \times 5$) Write your prediction.
4. Empty the measuring cup, refill the drip cup, and run it for 5 minutes. Measure. Compare with your prediction.
5. Watch the drips near the end. Are they as fast as at the start?

What you should see: The 5-minute total comes out LESS than $\text{rate} \times 5$. The drip slows as the cup empties (less water pushing on the hole — less pressure, from last week's Physics). The rate was a falling curve, not a flat line, so the true total is the area under that curve — smaller than the rectangle. This is exactly why calculus needs thin slices.